

Individual Assignment

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1. Introduction

The goal of this assignment is to design an interactive, coordinated visualization for the Housing dataset. The system consists of two linked views: (1) a scatterplot showing the relationship between house area and price, with a 2D rectangular brush, and (2) a strip-plot showing all houses positioned along the number of bedrooms, with a 1D horizontal brush. Selections made in one view are synchronized with the other: brushing or clicking in one visualization highlights the same subset of houses in the second visualization. In this report I explain how the visual encodings fit the properties of the data and the user tasks, and discuss the main advantages and limitations of my design choices.

2. Data and user tasks

The Housing dataset contains quantitative attributes (for example price and area) and more categorical or low-cardinality attributes (for example bedrooms). Typical analysis questions for this type of dataset are: how does price relate to house size, are there clusters or outliers, and how does this relationship change for houses with a different number of bedrooms. Therefore, the main user tasks I target are: overview (get a global impression of the distribution of houses), explore relations (see correlation and patterns between area and price), subset and compare (select a subset, such as houses with three bedrooms, and compare it to the rest), and identify outliers (highlight unusual points, such as very small but expensive houses).

3. Scatterplot: area vs. price with 2D brush

In the first view, the x-axis encodes area and the y-axis encodes price. Each house is represented as a circle positioned by these two values. All points are drawn with low opacity, while selected points are rendered with higher opacity and a thicker red stroke to stand out. This encoding directly matches the nature of the attributes: both variables are numeric and it is natural to represent their joint distribution as a scatterplot. The viewer can immediately assess whether there is a positive relationship between area and price and how strong it is, as well as see regions with higher point density.

The scatterplot supports two interaction techniques: click selection and 2D rectangular brushing. Clicking a single circle selects exactly one house. This is useful for inspecting a specific item while seeing where it lies in both views. The rectangular brush allows the user to drag a rectangle over any region of the scatterplot to select all houses in that area. This supports tasks such as selecting large and cheap houses or very expensive small houses. The interaction directly matches the analytic tasks, because the user can define subsets based on both variables simultaneously.

4. Second view: bedrooms strip-plot with 1D brush

The second view is a one-dimensional strip-plot based on the bedrooms attribute. The x-axis encodes the discrete values of bedrooms using a band scale. The y-position of each point is a small random jitter within the vertical range, to avoid overplotting while keeping the plot visually simple. Each house is again a small circle; selected houses are highlighted with

higher opacity and a thicker red outline. This encoding treats bedrooms as a small categorical or ordinal variable, which is appropriate because it has few distinct integer values. The strip-plot shows how many houses fall into each bedroom category and which of those belong to the currently selected subset.

Interaction in this view mirrors the scatterplot. A click selects a single house and highlights it in both views. A 1D brush along the x-axis allows the user to select an interval of bedroom counts (for example, all houses with two to three bedrooms). All points whose x-position falls inside the brush are selected. This view focuses on categorical comparison rather than continuous correlation, and is particularly useful for questions such as how the subset of large, cheap houses is distributed across bedroom counts.

5. Coordinated multiple views

The two views are tightly linked through selection. When the user brushes or clicks in the scatterplot, the selected subset is passed to the second view and highlighted there. When the user brushes or clicks in the bedroom view, the same subset is highlighted back in the scatterplot. This design follows the coordinated multiple views pattern: one view is well suited for exploring numerical relationships, while the other is good for inspecting how the same subset is distributed across a categorical variable. The coordination reduces cognitive load because the user does not need to remember filters or manually replicate them.

6. Discussion of design choices

There are several advantages to this design. First, the match between data type and visual encoding is good: scatterplots are standard for analysing relationships between two quantitative variables, and the strip-plot is effective for a low-cardinality numeric or categorical variable such as bedrooms. Second, the combination of a 2D brush in the scatterplot and a 1D brush in the strip-plot provides expressive selections and supports rich subset definition. Third, highlighting with opacity and stroke width is clear and consistent across both views. Fourth, the visualizations scale reasonably well for moderate data sizes, as points are small and semi-transparent. Finally, the interactions are simple and familiar, based on brushing and clicking.

The design also has some limitations. With a very large number of houses, both plots would suffer from overplotting; a density or aggregation-based visualization could be more appropriate in that case. Only a few attributes are shown explicitly, so understanding the influence of other variables would require additional encodings or views. In the strip-plot, the vertical jitter has no semantic meaning, which users must be aware of. Finally, the design emphasises visual patterns rather than precise numeric values, so additional views such as tooltips or details-on-demand panels would be needed for exact readings.

7. Conclusion

The final design combines a scatterplot of area versus price with a strip-plot over the number of bedrooms, linked through interactive brushing and clicking. The chosen visual encodings match the data types and support the main analysis tasks: exploring relationships, selecting meaningful subsets, and comparing them across attributes. Despite some limitations, the system provides a clear, simple and effective way to interactively explore the Housing dataset and to understand how its main attributes relate to each other.